


Households supply labor (N)

$$U(c, l)$$

$$\text{Hours} = h = N + l$$

* no future (no savings)

* no past (no K)

Graph trick: plot points
of equal utility: $U_1 = U_2$

points along this plot

IC = Indifference Curve

* tradeoff: less leisure

$\Rightarrow$ more work

$\Rightarrow$ more GDP

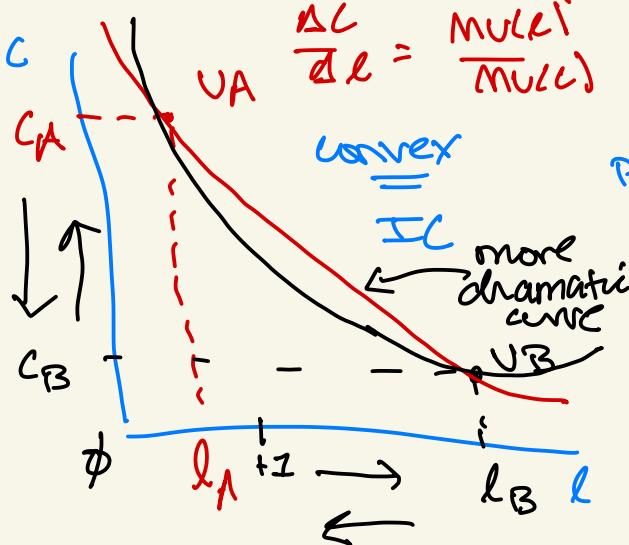
and more C

$$l \downarrow \quad U \downarrow$$

$$C \uparrow \quad U \uparrow$$

$$GDP = Y = C$$

Rise over Run = slope of IC



U assumptions
1. More is preferred to less

$\Rightarrow$ gives us downward-sloping IC

assume $l \uparrow$
need $C \downarrow$ to be on same IC

2. Diversity has value
as $l \uparrow$ $MU(l) \downarrow$ } diminishing returns to l
marginal utility of l

as $C \uparrow$ $MU(C) \downarrow$ } diminishing returns to C

From l_A to l_{A+1} } $\frac{MU(l) \text{ big at } A}{MU(l) \text{ small at } A} = \frac{\Delta C}{\Delta l}$
big ratio at point A

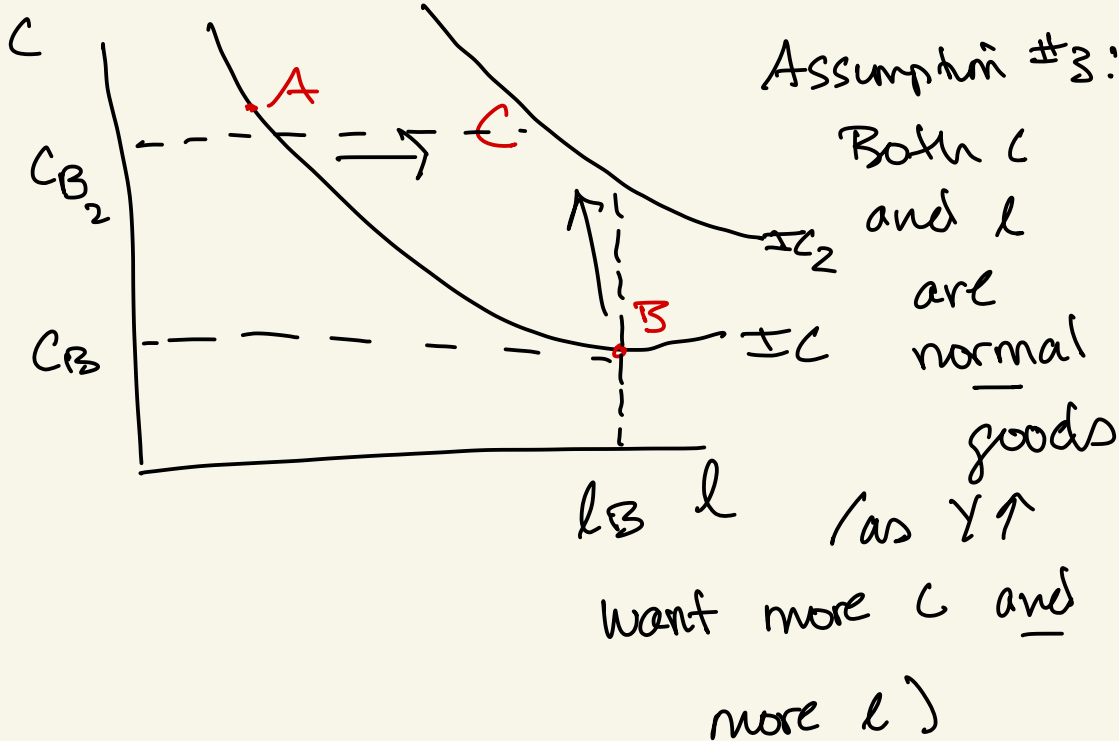
ΔC = # of C -goods I'd give up to get $MU(l)$ (price of l in terms of C)

convexity

foregone

IC flatter as $l \uparrow$ ($MU(l) \downarrow$)

slope = $\frac{MU(l)}{MU(C)} \downarrow$ as $l \uparrow$



* If $Y \uparrow$ then IC shifts $\uparrow$ because you can't have both $C \uparrow$ and $l \uparrow$ on same IC

* C is a point where Y must be higher

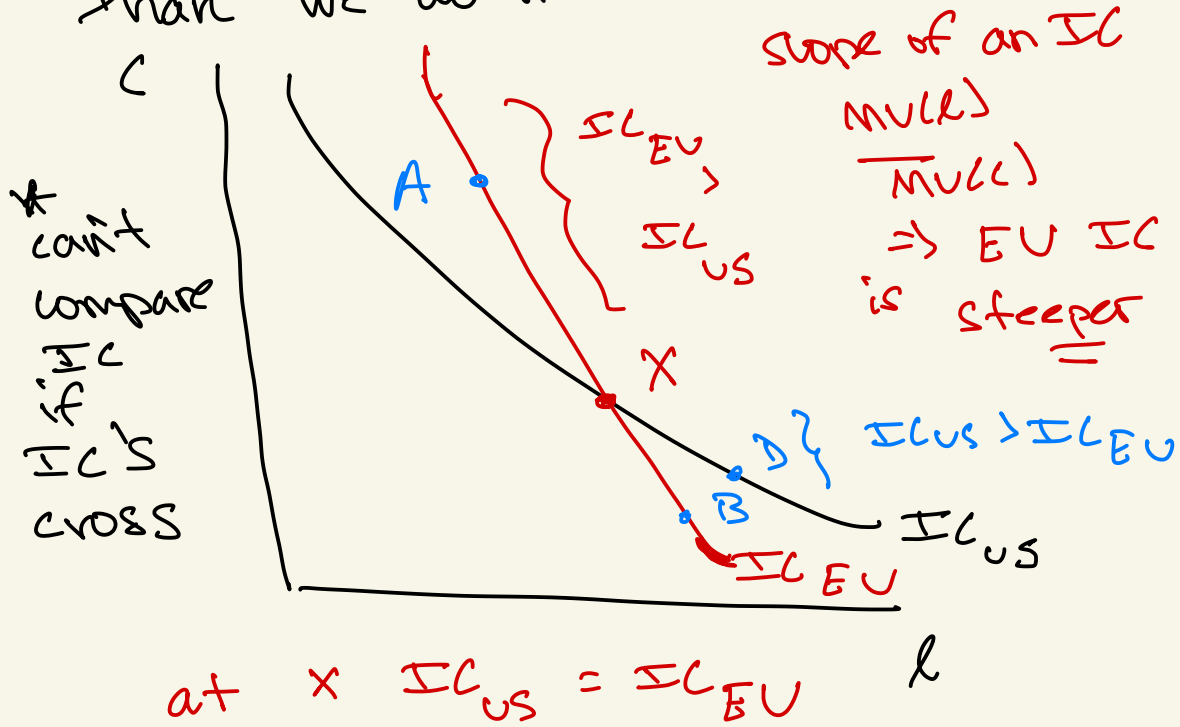
$$IC_2 \succ IC_1$$

IC 's are rankable

Holding l constant...
if at l_B $C_B \uparrow$ only if income $\uparrow$

"representative agent"
 $\Rightarrow$ everyone has same IC
 within a country

Compare US and EU assuming
 people in EU value leisure more
 than we do in the US



$$U_A = U_X = U_D \\
> U_B = U_X$$

example

$$U[l, c] = a \ln c + b \ln l$$

$a, b > 0$ } more preferred to less

$$\frac{MU(l)}{MU(c)} = \frac{\frac{b}{l}}{a/c} = \frac{b}{l} \cdot \frac{c}{a}$$

$$\text{slope} = \frac{c}{l} \cdot \frac{b}{a}$$

as $l \uparrow$ slope $\downarrow$

we have convexity



diversity has value
IC is convex

Household Income ? Budget constraint

$$C = W(1-t)N + \pi - T$$

W = real wage

t = proportional income tax

N = hours worked

(t is pure waste)

π = profits & nonwage income

T = lump sum tax paid even if don't work

Assume $\pi > T$

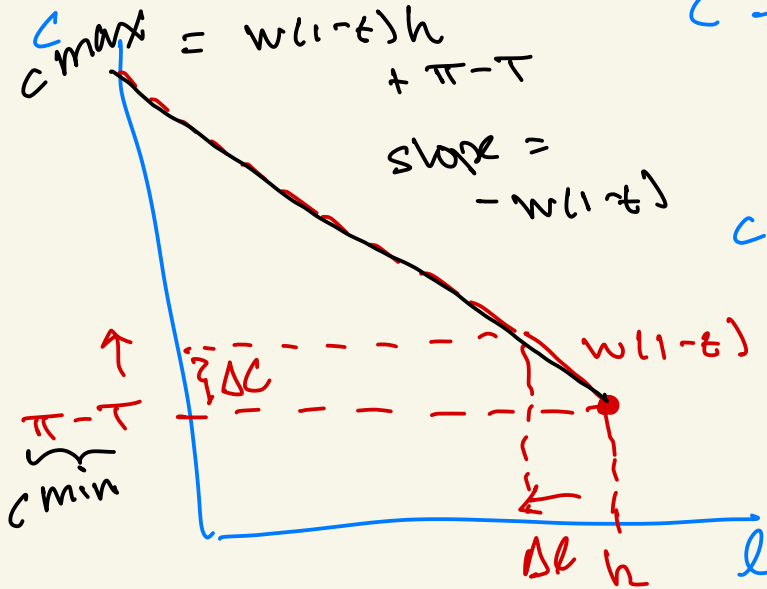
$$C = w(1-t)N + \pi - T$$

$$h = N + l$$

$$h - l = N$$

$$C = w(1-t)(h-l) + \pi - T$$

$$= w(1-t)h - w(1-t)l + \pi - T$$



if $l = h$ ($N=0$)

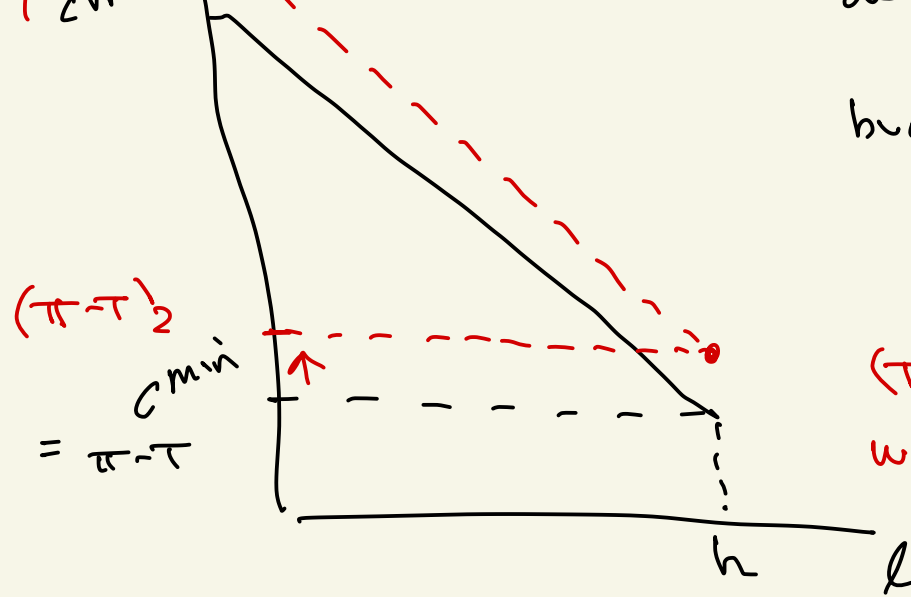
$$C = \pi - T \quad (N=0)$$

as $l \downarrow$ $N \uparrow$ start earning at rate $w(1-t)$ $\{ C \uparrow$ as $l \downarrow$
slope $-w(1-t)$

Assume $\pi > T$

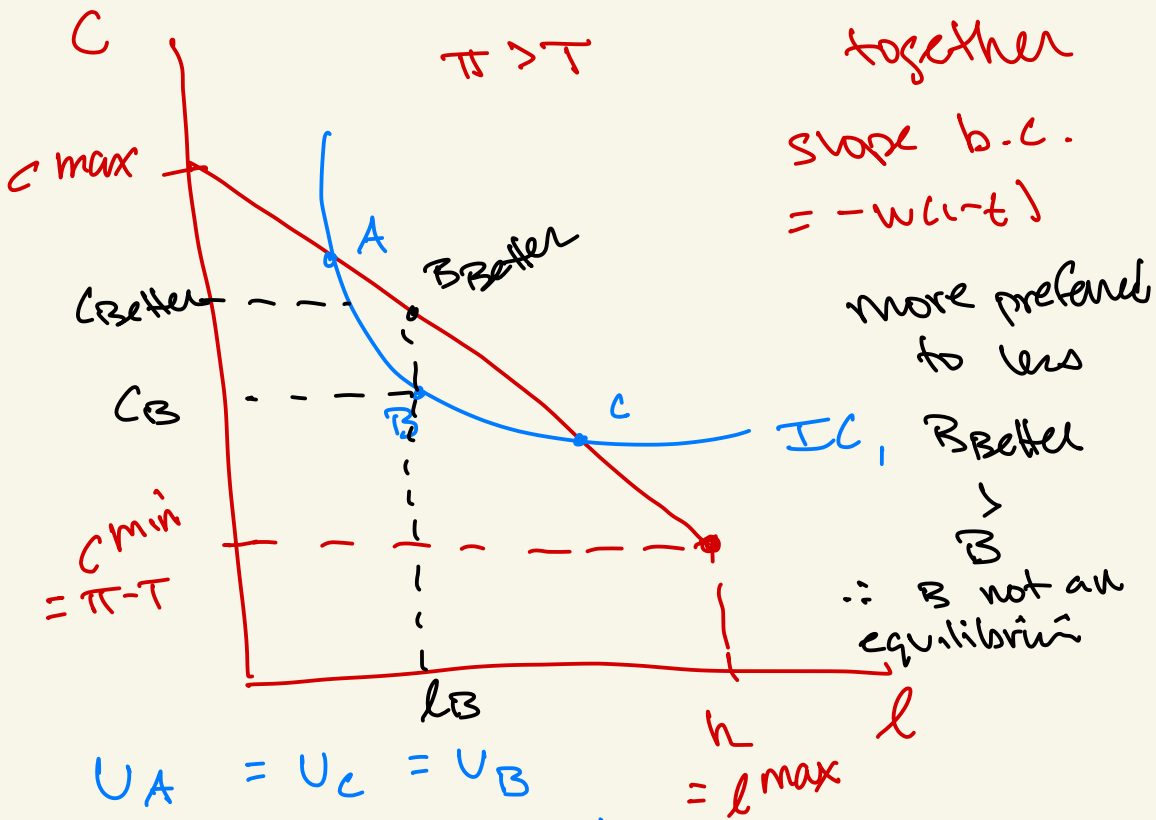
$$C^{\max}_2 = w(l+\pi)h + (\pi-T)l_2$$

$$C^{\max} = w(l+\pi)h + \pi - T$$



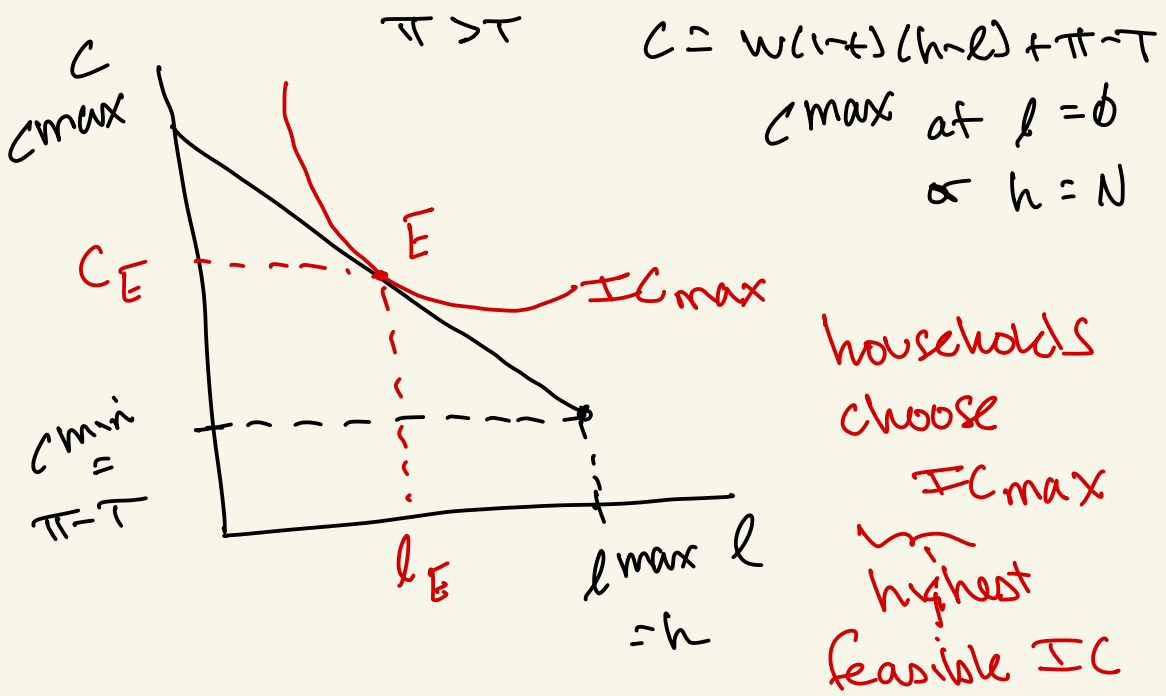
as $\pi \uparrow$
 $\sigma T \downarrow$
 budget
 constraint
 shifts
 out
 $(\pi - T) \uparrow$
 with no
 Δ slope

Putting IC and budget constraint together



But B is inside my budget constraint and we assume we maximize utility

if B no good = A and C are no good



To find E :

Set slope of $IC =$ slope of

the budget constraint
along the budget constraint

$$\frac{+ \frac{MU(l)}{MU(c)}}{+ w(1-t)}$$

marginal rate of substitution
 $= MRS$